

END-TO-END PIPELINE & COUPLING REPORT

NGROK → GITHUB → CLOUDFLARE DEPLOYMENT INFRASTRUCTURE

Project Name: messyMesh Archive System

Root Folder: ~/Documents/messyMesh

Status: Live / Operational

1. The Complete Deployment & Automation Pipeline

The messyMesh application relies on a continuous deployment (CD) architecture. Changes flow seamlessly from your local computer, through cloud source control, and straight out onto edge hosting servers:

Step 1: Local Engineering Workspace → You modify code components (like `App.jsx` or `Dashboard.jsx`) inside your project directory at `~/Documents/messyMesh` using your code editor.

Step 2: Source Tracking Verification → Running `git add .` and `git commit` snapshots these configurations locally inside the local hidden tracking layer.

Step 3: GitHub Push Sync → Running `git push origin main` broadcasts these files straight to your remote cloud repository at `github.com/kcplibrary/messhyMesh_repo`.

Step 4: Cloudflare Automated Compilation → Cloudflare Pages intercepts the incoming GitHub commit message, provisions a background build server, wraps up your asset assets into static bundles, and re-deploys your frontend globally to your custom domain: `repo.kcplibrary.com`.

2. Live Data Traffic Pipeline (The Dynamic Bridge)

Once loaded onto a terminal device or mobile smartphone browser via Cloudflare, live request queries route down a safe, custom data bridge back into your hard drive:

1. Frontend Initialization: The phone opens `repo.kcplibrary.com`. When you click "Login" or look at data loops, the browser fires an HTTP asynchronous request.

2. Public Proxy Ingress: The request hits your public ngrok gateway address (e.g., `https://customer-yahoo-outing.ngrok-free.dev`).

3. Security Token Exemption: The custom Axios header parameter (`ngrok-skip-browser-warning: true`) hits the gate, preventing ngrok from popping up an interstitial validation screen.

4. Internal Translation: The ngrok agent running on your desktop translates this traffic down to local interface address `http://127.0.0.1:8000`.

5. Path Execution & Target: Traffic points directly to the server root at `/backend/api/...` to execute data reads and database lookups safely.

3. Breakdown of Recent System Updates

CRITICAL STRUCTURAL UPDATES IMPLEMENTED:

- **Path Alignment:** Frontend paths were updated from localized tracking arrays (/api/login.php) to explicit, absolute file system directories (/backend/api/login.php). This instantly stopped 404 No Such File or Directory server errors.
- **CORS Handshake Injections:** Wrote Access-Control-Allow-Origin: * scripts directly onto the absolute top of the database configuration initialization layer (db_connect.php). This dropped browser security blocks across the network.
- **Unified Success Listeners:** Streamlined cross-language response schemas. The PHP backend responds with clean JSON statuses ("status": "success"), which are caught uniformly by frontend conditional logic blocks to write local session state keys (mesh_session) instantly.

4. Daily Development & Ingress Checklist

Use these steps when starting your workspace or publishing layout tweaks:

```
# Tab 1: Initialize Local Server Instance
cd ~/Documents/messyMesh
php -S 127.0.0.1:8000

# Tab 2: Expose Secure Pipeline
ngrok http 8000

# Tab 3: Publish Tweaks and Trigger Automatic Cloudflare Rebuilds
git add .
git commit -m "Synchronizing latest environment assets"
git push origin main
```